



Antonis Papaioannou
Senior Database Researcher
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SUMMARY

Senior Database Researcher with 10+ years of experience in data management systems and distributed systems. Currently developing GaussDB at Huawei Research, focusing on cloud-native analytics and HTAP workloads. PhD in Computer Science with 20+ peer-reviewed publications and a track record of research recognition, including **selection as a finalist for the 2022 ERCIM Cor Baayen Young Researcher Award** and Huawei's **Outstanding Contributor Award, Future Star Award** and **Presidents Award**. Proven track record bridging academic research with industrial applications in systems like Apache Cassandra, MongoDB, Redis, Apache Flink and RocksDB.

AWARDS AND RECOGNITION

- **Outstanding Contributor Award**, Huawei Research, 2025
- **Presidents Award - Significant Business Contribution**, Huawei Research, 2024
- **Future Star Award**, Huawei Research, 2023 and 2025
- **Database Key Technology Breakthrough Team Award**, Huawei Research, 2023
- **Finalist, ERCIM Cor Baayen Young Researcher Award, 2022** — selected among eight finalists; nominated by Prof. Dimitris Plexousakis (ICS-FORTH)
- Grand Challenge Award, ACM DEBS, 2022
- Best Poster Award, ACM EuroSys, 2022
- Recipient of multiple competitive fellowships including the Hellenic Foundation for Research and Innovation PhD Scholarship and Maria Michail Manasaki Doctoral Fellowship.

PROFESSIONAL EXPERIENCE

SENIOR DATABASE RESEARCHER

Sep 2022 – Present

HUAWEI RESEARCH, EDINBURGH, UK

Developing core GaussDB features serving enterprise and Cloud clients globally:

- Designed and implemented query segmentation improving scheduling and resource efficiency in the cloud-native analytics platform.
- Integrated Iterative Dynamic Programming (IDP) into the query optimizer boosting performance for complex queries.
- Led the development of core features for Incremental View Maintenance (IVM) to enable low-latency, delta-based updates to materialized views.
- Recognized with Future Star Award (2023) and contributed to two team breakthrough awards

POSTDOCTORAL RESEARCHER

Dec 2021 – Aug 2022

INFORMATION SYSTEMS LABORATORY (ISL), ICS-FORTH

- Designed adaptation techniques in large-scale distributed data stores and stream-processing systems.
- Developed elasticity and performance optimization mechanisms used in Apache Cassandra, MongoDB, Redis, RocksDB, Apache Flink, and Apache Samza.

RESEARCH ASSISTANT

Nov 2011 – Nov 2021

ICS-FORTH

- *Information Systems Laboratory (ISL)*, Nov. 2016 – Nov. 2021: Designed performance and scalability improvements for distributed storage systems.
- *Computer Architecture and VLSI Systems (CARV) Laboratory*, Nov. 2012 – Oct. 2013: Designed and implemented frameworks for evaluating distributed application deployments in multi-cloud platforms; deployed and maintained backend database infrastructure for cloud-based social networks.
- *Distributed Computing Systems (DCS) Laboratory*, Nov. 2011 – Oct. 2012: Studied the impact of social media on website performance and security.

RESEARCH ASSOCIATE

Nov 2013 – Nov 2014

COMPUTER ARCHITECTURE AND VLSI SYSTEMS (CARV) LABORATORY, ICS - FORTH

Developed a platform for resource performance classification in the Cloud; identify performance improvements in multi-Cloud deployments leveraging execution history.

SYSTEMS & NETWORK ADMINISTRATOR

Mar 2018 – Dec 2018

HELLENIC ARMY, RESEARCH AND INFORMATION CORPS

Managed critical infrastructure hosting military applications; configured Dell PowerEdge VRTX servers; redesigned secure network topology for classified information systems

SOFTWARE DEVELOPER & DATABASE ADMINISTRATOR

Apr 2012 – Sep 2012

HELP AT HOME, HERAKLION, CRETE, GREECE

Designed customer data models and developed employee task assignment application

DATABASE ADMINISTRATOR

Sept 2009 – Apr 2010

HANAPTIXI, HERAKLION, CRETE, GREECE

Designed customer data models and developed employee task assignment application. Responsible for database infrastructure, maintenance, tuning, data backup and replication

SELECTED RESEARCH PROJECTS

- StreamStore (2020–2021) – Elastic, scalable, fault-tolerant, stateful stream-processing platform.
- EVOLVE (2019–2021) – HPC + Big Data integration for large-scale analytics in multiple domains.
- PaaSage (2012–2016) – Multi-cloud deployment & optimization framework.

SELECTED PUBLICATIONS

Selected publications from 20+ peer-reviewed papers:

- S. Kalogerakis, **A. Papaioannou**, and K. Magoutis, Efficient Processing of High-Volume Tick Data with Apache Flink for the DEBS 2022 Grand Challenge, in the *16th ACM International Conference on Distributed and Event-Based Systems*, 2022 (**Grand Challenge Audience Award**)
- **A. Papaioannou** and K. Magoutis, Amoeba: Aligning Stream Processing Operators with Externally-Managed State, in the *14th IEEE/ACM International Conference on Utility and Cloud Computing*, 2021
- **A. Papaioannou** and K. Magoutis, Addressing the read-performance impact of reconfigurations in replicated key-value stores, in the *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 2021
- **A. Papaioannou** and K. Magoutis, Replica-group leadership change as a performance enhancing mechanism in NoSQL data stores, *38th IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2018
- **A. Papaioannou** and K. Magoutis, Incremental elasticity for NoSQL data stores, in the *36th IEEE International Symposium on Reliable Distributed Systems (SRDS)*, 2017
- **A. Papaioannou** and K. Magoutis, Streamstore: Aligning Stream-processing Systems with Externally-Managed State, in the *7th ACM SIGOPS European Conference on Computer Systems (EuroSys'22)*, Rennes, France, April 2022. **Best poster award!**

PROFESSIONAL ACTIVITIES

- Organizer, "Next-Generation Data Management Systems" industry event, EDBT 2025, Barcelona
- Shadow Program Committee Member, EuroSys 2020
- Lead Organizer, Career Fair events, University of Crete, 2021
- Reviewer: DAIS 2022, CCGrid 2014, Euro-Par 2013

TEACHING EXPERIENCE

TEACHING ASSISTANT, COMPUTER SCIENCE DEPARTMENT, UNIVERSITY OF CRETE

2010-2021

Key courses: Operating Systems, Modern Topics in Scalable Storage Systems, Data Structures, Computer Networks, Web Programming

EDUCATION

PHD IN COMPUTER SCIENCE UNIVERSITY OF CRETE, GREECE

Nov 2021

Dissertation: Efficient adaptation mechanisms for distributed data stores

Advisor: Prof. Kostas Magoutis

M.Sc. IN COMPUTER SCIENCE UNIVERSITY OF CRETE, GREECE

Nov 2013

GPA: 9.21/10.0

Major: Parallel and Distributed Systems, Minor: Information Systems

B.Sc. IN COMPUTER SCIENCE UNIVERSITY OF CRETE, GREECE

Jun 2010

GPA: 7.21/10.0